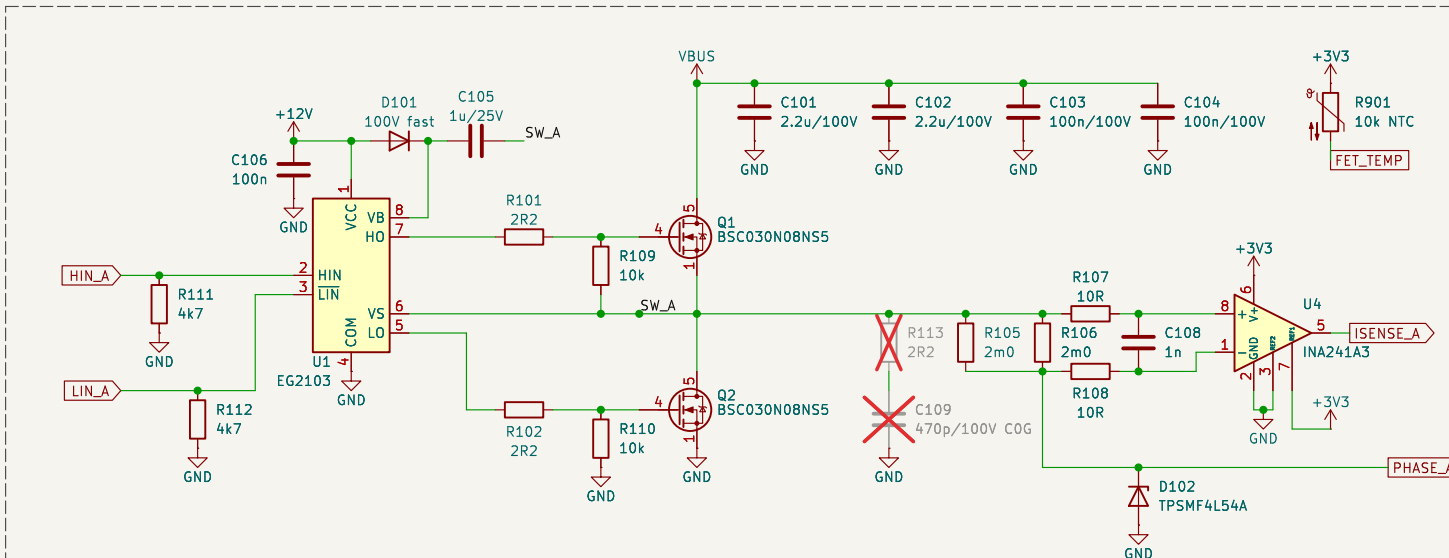
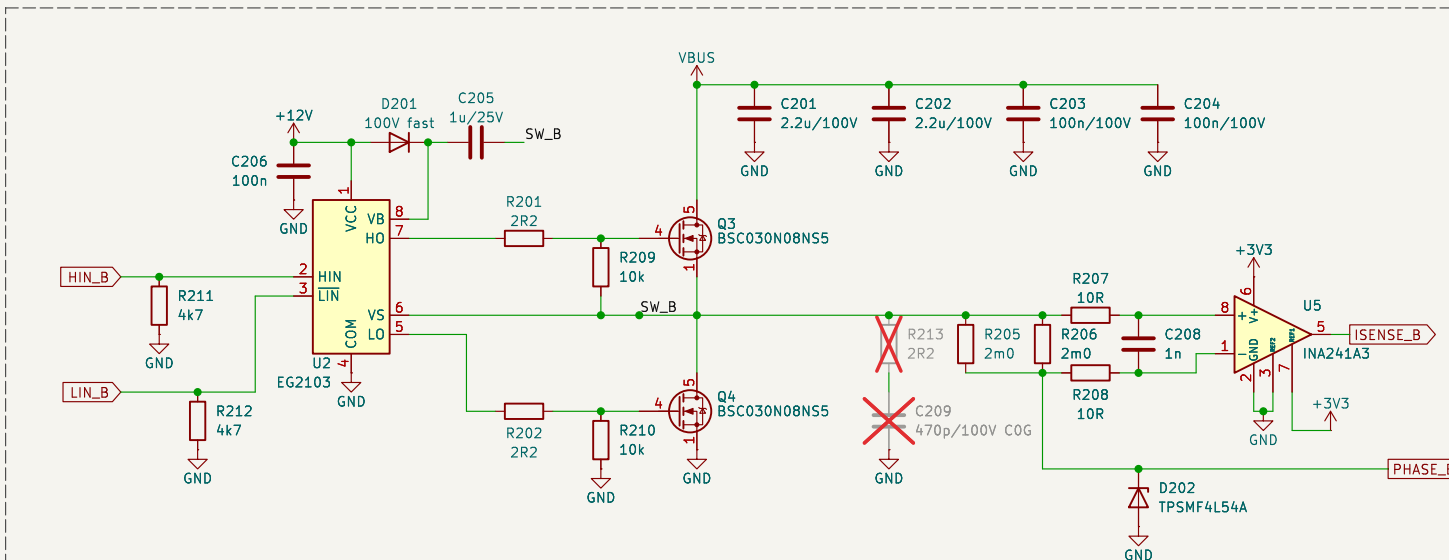


	1	2	3	4	5	6
	Board A plus the minimum of board B. 48 V max. XT30 bus input, TVS and cans, two LMR38010 rails, USB-C, RS-485 relay on two SH 6, expansion header.					
A	01_power_stage Three half-bridges: 6x BSC030N08NS5, 3x EG2103, bootstrap, gate networks, 2x inline shunt + INA241A3, DC-link ceramics File: 01_power_stage.kicad_sch 02_control RP2350A QFN-60, crystal, QSPI flash, 3V3 LDO, ADC supply File: 02_control.kicad_sch 03_encoder MT6701 on the motor-facing side, SSI to SPI0 File: 03_encoder.kicad_sch 04_power XT30 bus input, SMDJ54A, 2x 100 uF polymer, bus divider, 2x LMR38010: 12 V gate rail (GATE_OFF stops it), 5 V logic File: 04_power.kicad_sch 05_io USB-C (data + 5 V), RS-485 relay on two SH 6, expansion header, RGB LED, BOOTSEL, test pads File: 05_io.kicad_sch					A
B						B
C						C
D	CERN-OHL-P See ../../single.html Sheets drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py Sequoia Hope Alexander Sheet: / File: servodrive_S.kicad_sch Title: servodrive – board S, single board Size: A4Date: 2026-09-07 KiCad E.D.A. 9.0.8Rev: A Id: 1/6					D
	1	2	3	4	5	6

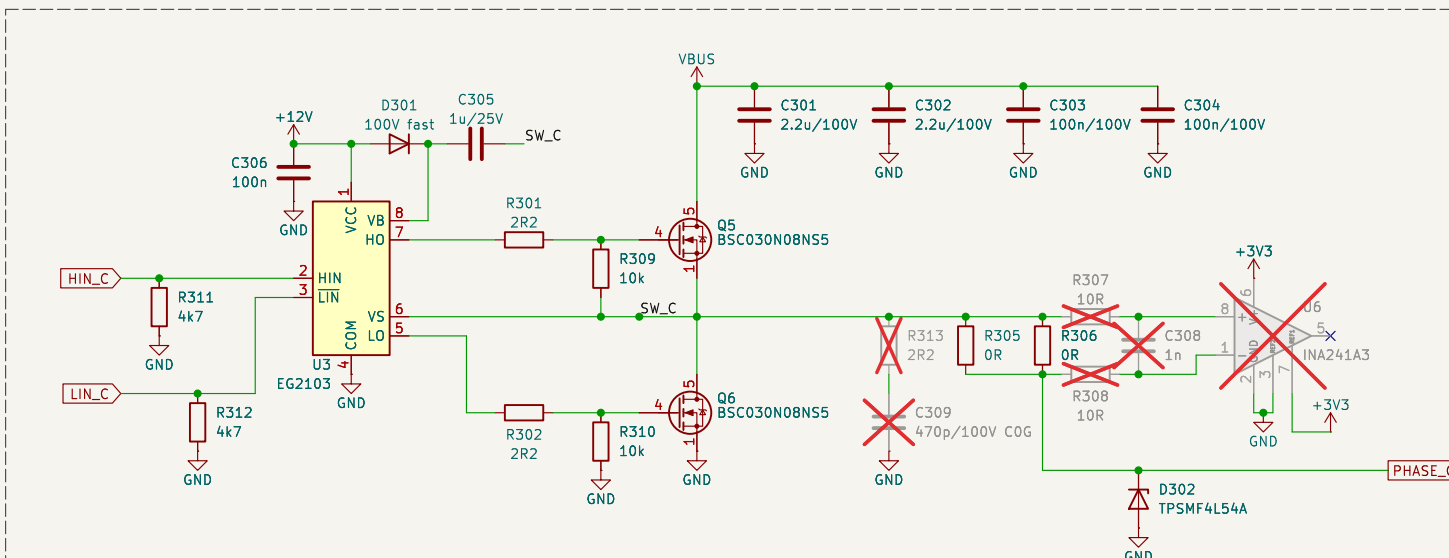
Phase A



Phase B



Phase C



Notes

Each phase is an EG2103 half-bridge driver on a pair of BSC030N08NS5: bootstrap diode and capacitor (D101, C105 on phase A), 2R2 gate resistors, 10k gate-source pull-downs.

HIN and LIN are held low (R111, R112): with both low the EG2103 turns the low side on, so a CPU in reset brakes the motor.

Inline shunt: two 2mΩ in parallel between the switch node and the motor lead, Kelvin-sensed through 10R, 10R and 1n into an INA241A3. REF1 on +3V3 and REF2 on GND put its output at mid-rail, so it reads current either way.

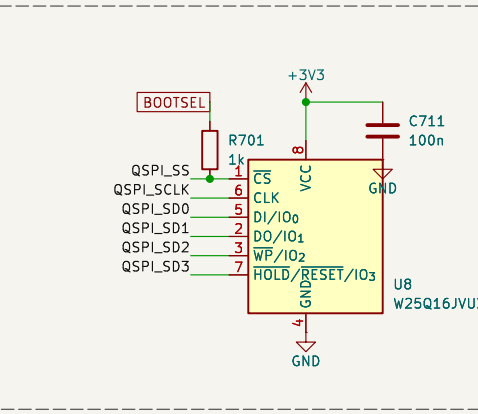
Phase C: the shunts are OR links and U6, R307, R308, C308 are not fitted. The RP2350A has no fourth ADC channel; phase C current is $-(I_a + I_b)$.

R113 + C109 (and B, C): RC snubber across each low side, footprints only.

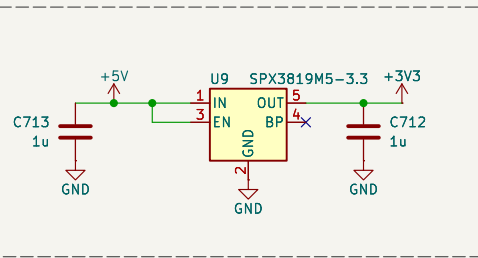
R901: NTC beside the FETs, read on ADC3 as FET_TEMP.

CERN-OHL-P		
Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py		
Sequoia Hope Alexander		
Sheet: /01_power_stage/		
File: 01_power_stage.kicad_sch		
Title: servodrive – board S, single board – 01_power_stage		
Size: A3	Date: 2026-09-25	Rev: A
KiCad E.D.A. 9.0.8		Id: 2/6

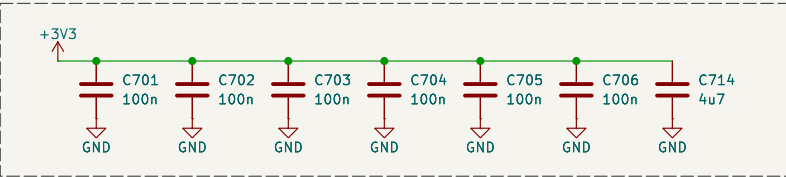
QSPI flash



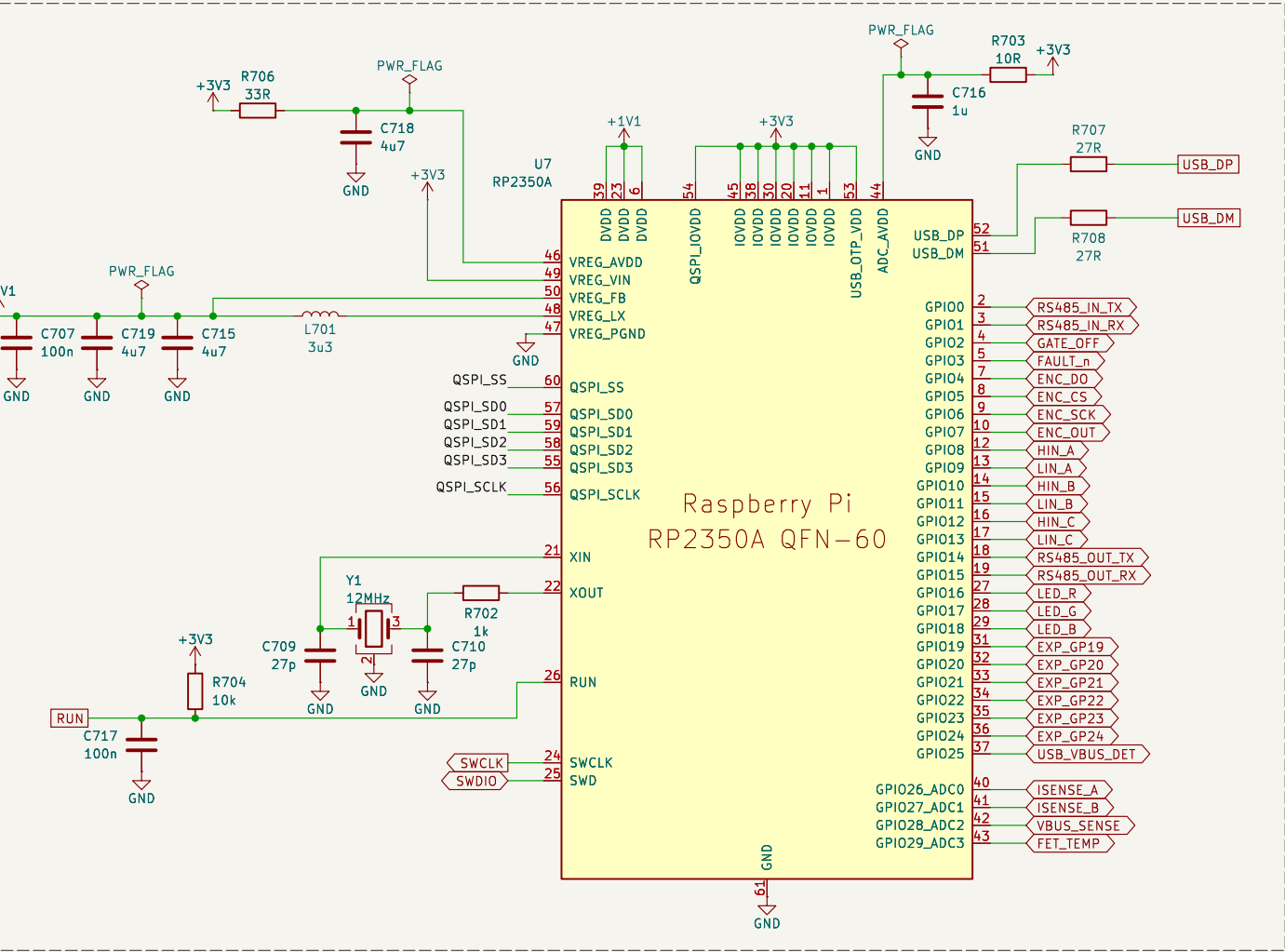
3V3 LDO



Decoupling



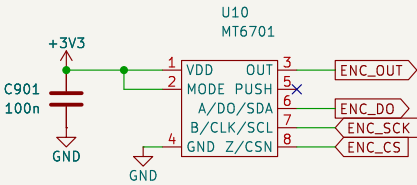
RP2350A



Notes

- RP2350A as Raspberry Pi's minimal design (RP-006440): the core supply is the chip's own switching regulator, VREG_LX through L701 to +1V1, which VREG_FB senses and DVDD takes. VREG_AVDD is +3V3 through R706 and C718.
- One 100n at each IOVDD pin (C701–C706); C714 4u7 at VREG_VIN.
- ADC_AVDD is +3V3 filtered by R703 and C716 (+3V3A).
- BOOTSEL: SW1 on sheet 05 pulls QSPLSS low through R701 at reset.
- The encoder: ENC_DO on GPIO4, ENC_SCK on GPIO6, ENC_CS on GPIO5, ENC_OUT on GPIO7 -- SPI0's RX and SCK, so SPI0 reads it.
- GPIO19–24 go to the expansion header (sheet 05).

Encoder



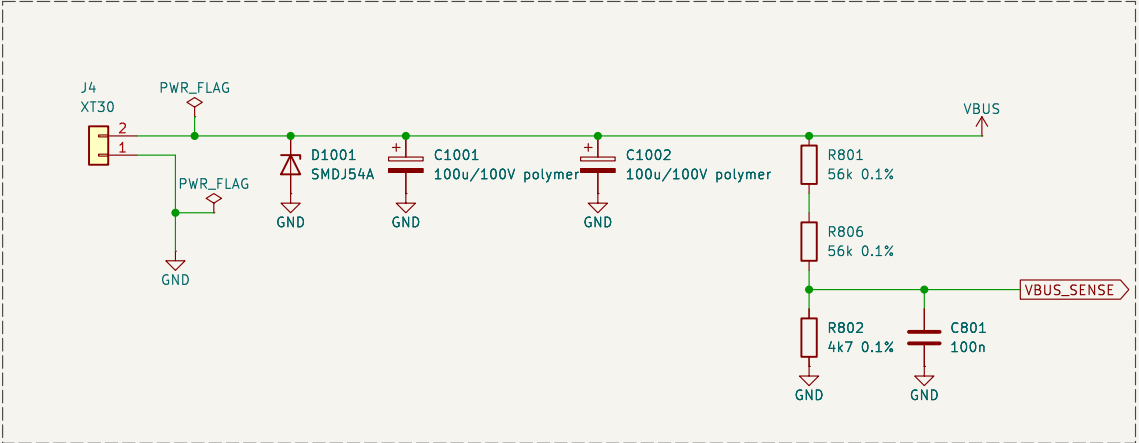
MT6701 on the motor-facing side, centred on the shaft over a diametric magnet. MODE tied to VDD. The CPU reads it over SSI (sheet 02).

CERN-OHL-P
Drawn by tools/schlayout_s.py from the netlist in tools/schematic_s.py
Sequoia Hope Alexander
Sheet: /03_encoder/
File: 03_encoder.kicad_sch

Title: servodrives – board S, single board – 03_encoder

Size: A4	Date: 2026-09-25	Rev: A
KiCad E.D.A. 9.0.8		Id: 4/6

Bus input



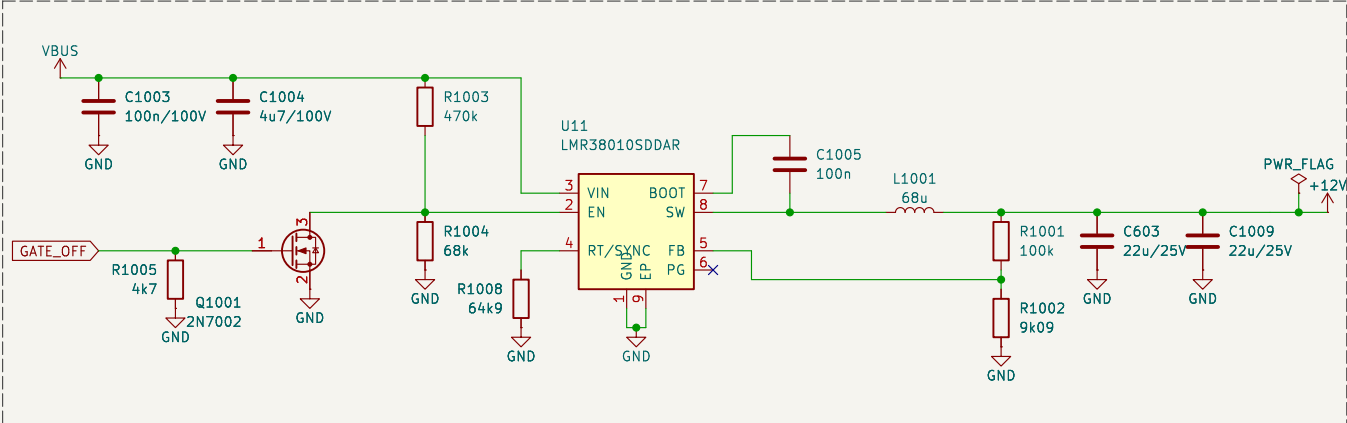
Notes

Bus: 48 V operating on 80 V FETs. D1001 (SMDJ54A, 54 V standoff) clamps it; C1001 and C1002 are the bulk.
 $VBUS_SENSE = VBUS \times 4k7 / 116k7$ (0.1 %), on ADC2.

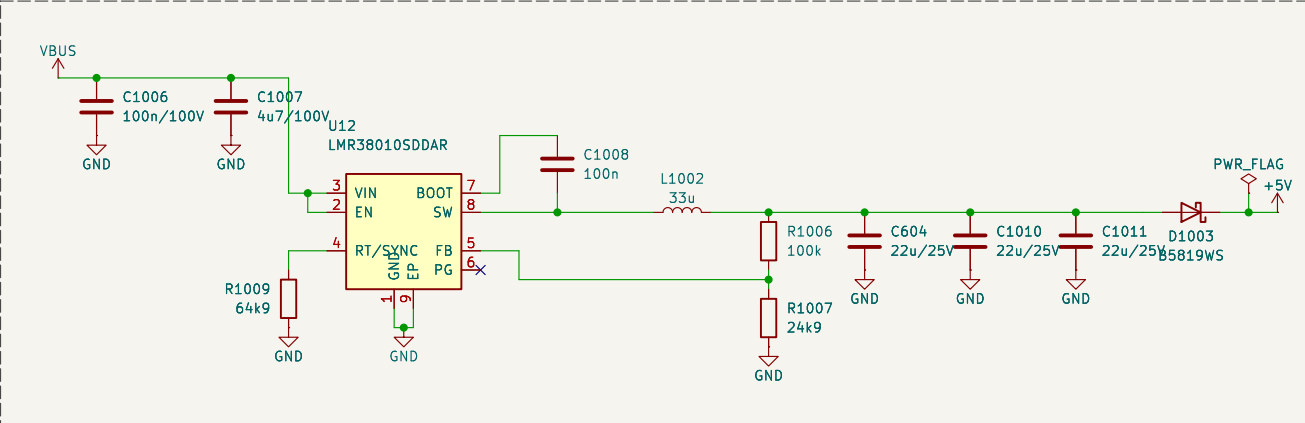
12 V gate rail: LMR38010, starting at 9.9 V on the R1003/R1004 divider. GATE_OFF (GPIO14) high turns Q1001 on and stops the rail, taking the gate drivers' supply away. R1005 holds it low through reset, so the rail comes up and the drivers brake the motor.

5 V logic rail: LMR38010, diode-ORed with USB (D1002, sheet 05) through D1003. The 3V3 LDO on sheet 02 runs from +5V.

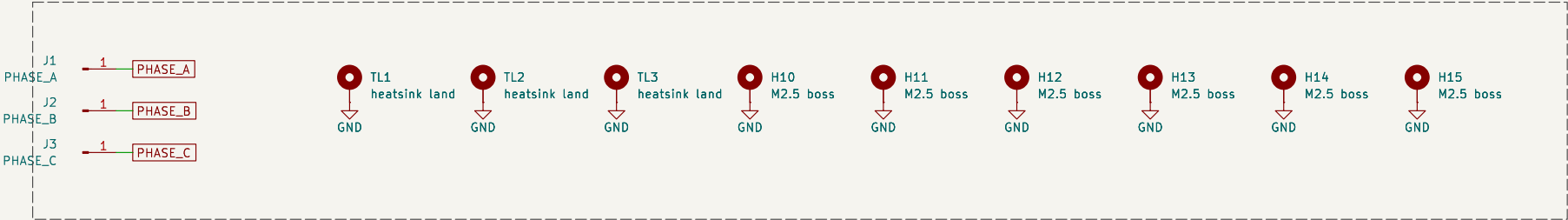
12 V gate rail



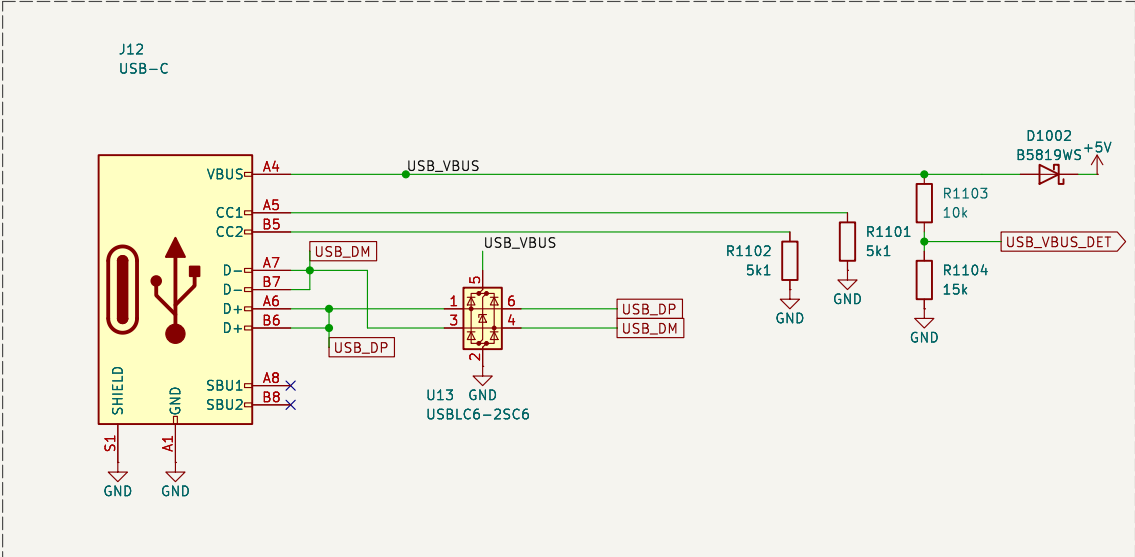
5 V logic rail



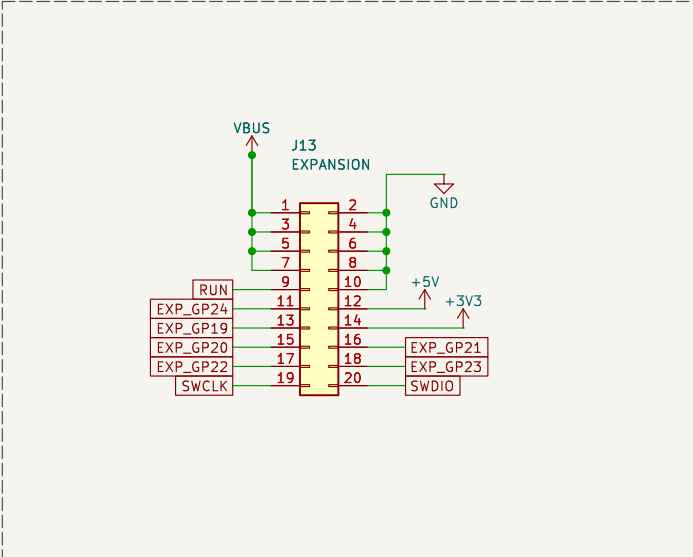
Mounting and motor leads



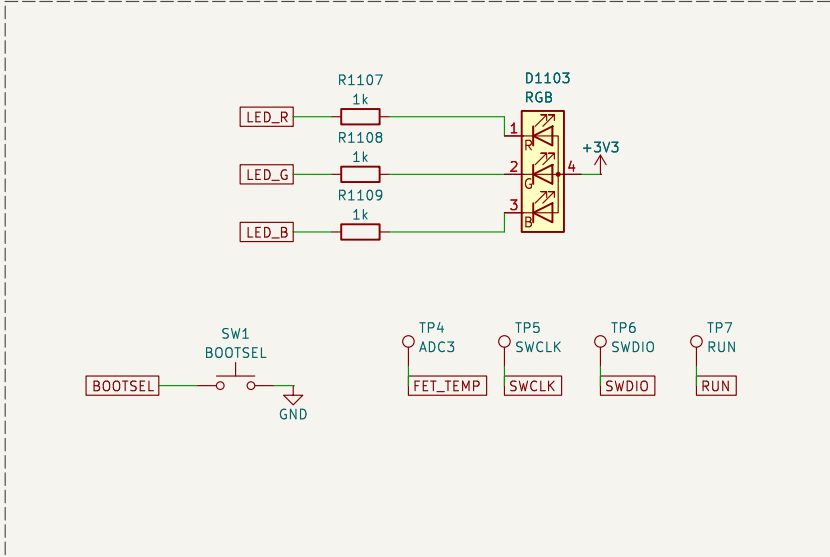
USB-C: data and 5 V logic, no PD



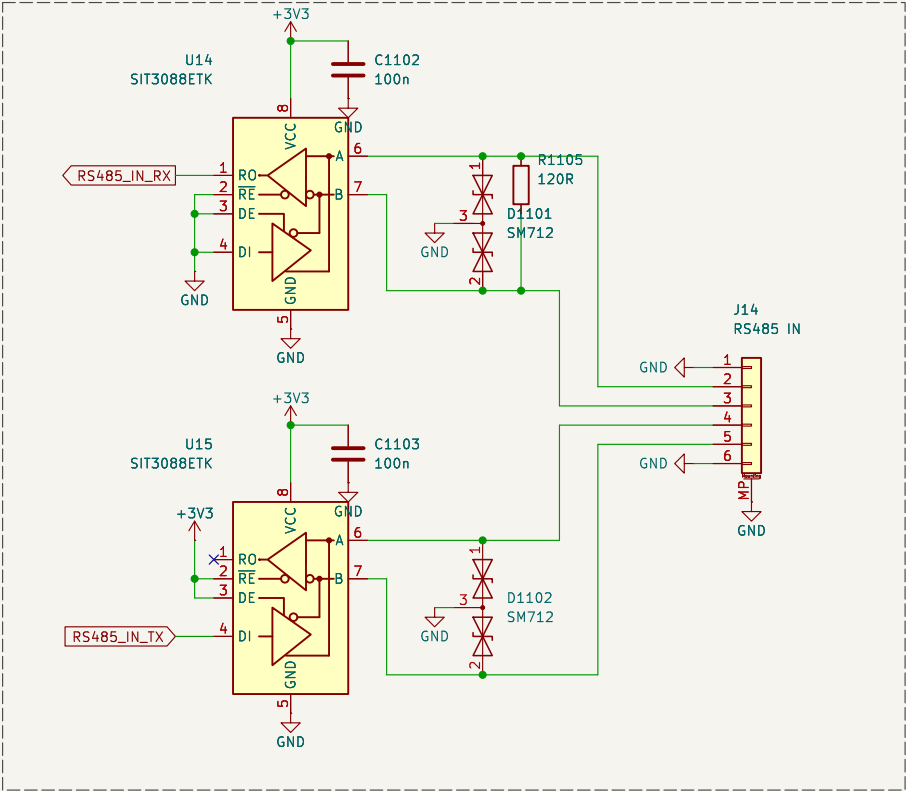
Expansion header: PD or Ethernet board



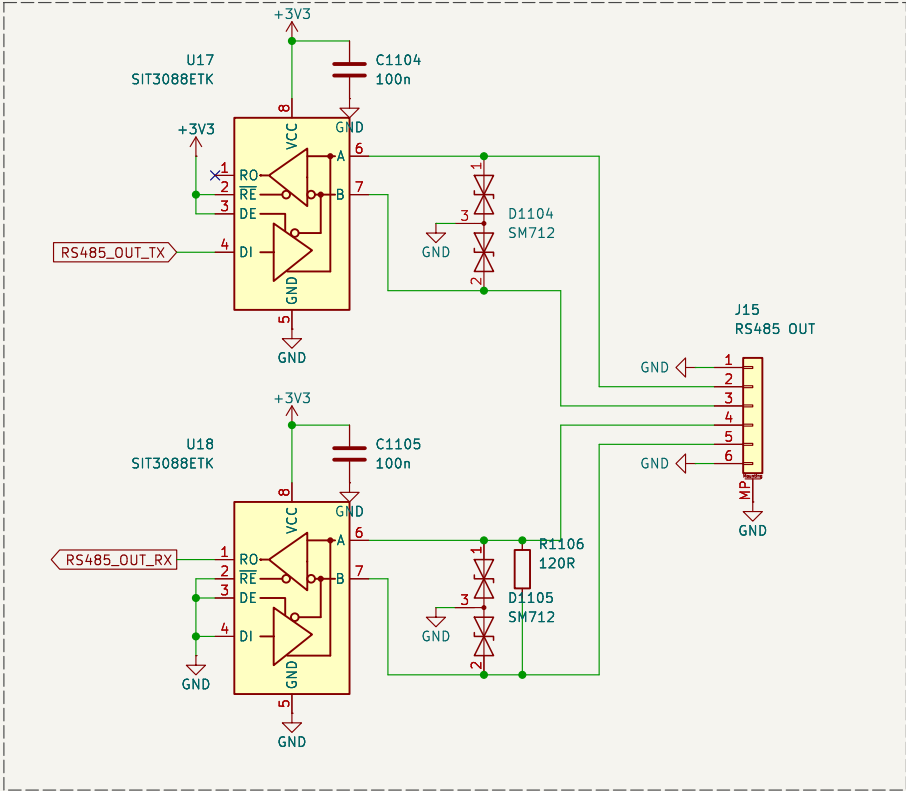
Status LED, BOOTSEL, test pads



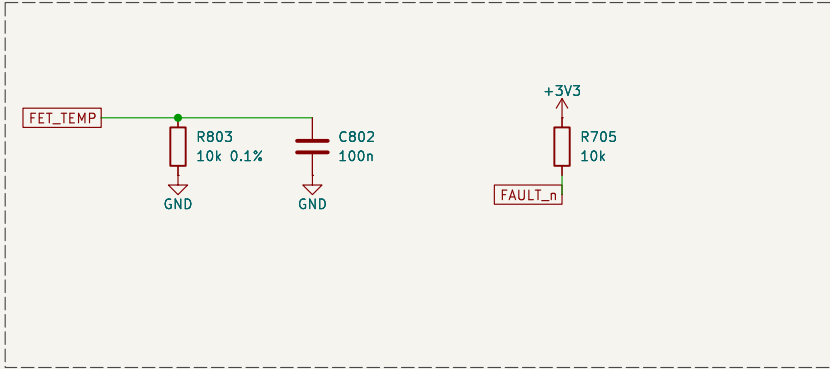
RS-485 relay: port IN



RS-485 relay: port OUT



ADC3 divider, FAULT_n pull-up



RS-485 is a CPU relay: each port is two point-to-point pairs, one in and one out, so a drive passes the chain on in software. Drivers are always enabled: each receiver has its 120R termination (R1105, R1106).

R803 with the NTC (R901, sheet 01) makes FET_TEMP.